

Explainable AI in Feature Selection: Improving Classification Performance on Imbalanced Datasets

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Abstract. Particularly in biomedical applications, feature selection plays a critical role in enhancing the interpretability and efficacy of machine learning models. This work examines the performance of Explainable AI (XAI), Information Gain (IG), and Principal Component Analysis (PCA) methods on an imbalanced dataset pertaining to stroke prediction. Data from 4,603 patient records, including 362 instances of stroke from the National Health and Nutrition Examination Survey are used in this investigation. Methodologically, IG is used for feature ranking, PCA is used to reduce dimensionality and XAI techniques are used to improve model transparency. The chosen features are used to assess the performance of several machine learning models, including Random Forest, Support Vector Machine, k-Nearest Neighbours, and Logistic Regression, in terms of classification. Our experimental results show that the combined PCA-IG approach significantly enhances classification accuracy, achieving 91.75%. Furthermore, LIME-based feature selection outperformed in precision, recall, and F1 score, with the highest accuracy at 91.86%. LIME discovered nine positive impact features, highlighting the top contributors in the dataset. We also applied the same feature selection technique to datasets from other domains. These findings highlight the robustness of using PCA-IG and XAI approaches separately to create reliable and understandable machine learning models for healthcare and other applications. By offering insights into the optimal use of PCA, IG, and XAI to enhance the accuracy and practicality of machine learning models in healthcare and other domains, this paper advances the field of feature selection across all areas of data analysis.

Keywords: Feature Selection · Classification · (Principal Component Analysis) · Imbalanced Data · Explainable AI.

1 Introduction

Data classification is the process of classifying data into distinct groups, which is often carried out in two stages: first, developing a model for the class attribute based on other factors, and then applying this model to fresh, previously unknown datasets to determine each record's class[5]. Classification algorithms, which are widely utilized in domains such as health, astronomy, business, biology, and media, frequently come across imbalanced datasets in which one class outnumbers

the other(s)[30,4,22]. This common occurrence in applications such as medical diagnosis, fraud detection, and spam filtering causes significant issues for machine learning algorithms, resulting in biased models that favor the majority while ignoring the minority[8,7,20].

Traditional feature selection approaches frequently face considerable hurdles when dealing with imbalanced datasets, owing to the overwhelming effect of the majority class [18]. This imbalance might distort the feature selection process, resulting in the identification of features that do not appropriately represent the minority class, leading to unsatisfactory and biased classification results. As a result, the model fails to generalize well, especially for the minority class, which is frequently the class of greatest interest in practical applications[11]. The problems of feature selection in imbalanced datasets are addressed in this work employing sophisticated approaches such as Principal Component Analysis (PCA), Information Gain, and Explainable AI (LIME). These algorithms are employed to identify and select important features, aiming to enhance classification performance[27].

We conducted a series of experiments on an imbalanced dataset using four classification algorithms: Random Forest Classifier (RFC), Support Vector Classifier (SVC), K-Nearest Neighbors (KNN), and Logistic Regressor (LR). Feature selection was performed using PCA, Information Gain, and LIME. After applying PCA, a feature selection algorithm was used. The models' performance was then evaluated in terms of accuracy, precision, recall, and F1 score.

This study highlights the importance of effective feature selection in improving the performance of machine learning models on imbalanced datasets. Using advanced techniques such as LIME, which assesses both positive and negative feature impacts, we show how reducing dimensions and selecting meaningful features can improve model accuracy and interpretability. The key contributions of this paper are:

- An extensive analysis of feature selection in the setting of imbalanced data using PCA, Information Gain, and LIME.
- A analysis of how each feature selection approach affects the performance of several classification algorithms.
- Evidence suggests LIME-based feature selection outperforms traditional approaches, especially in terms of accuracy and interpretability.

The rest of this paper is organized as follows: Section 2 provides a detailed review of related work. Section 3 discusses the dataset and experimental setup. Section 4 offers results and analyses. Section 5 explains the findings and their implications for society. Finally, Section 6 concludes the paper and suggests directions for future research.

2 Related Works

Working on any data mining or machine learning (ML) task necessitates increasingly effective strategies due to the sharp increases in data dimensions. Zebari et al. [32] focus on accuracy and dimensionality reduction as the primary goals of these methods. They highlight that redundant and unimportant features can hamper pattern classification. This research covers various datasets, including medical disease analysis, ethnicity identification, and emotion detection, employing feature selection (FS) and feature extraction (FE) methods. Their work on feature selection in power quality events outperformed other approaches, though there remains a risk of overfitting.

Khaire et al. [16] propose a novel stability estimator to address the instability of feature selection, emphasizing the importance of stability in feature selection methods across different domains.

They explore network algorithms for stable feature selection, such as RBNAs, and use the relative weighted consistency approach to tackle subset-size bias. They point out that instability arises when algorithms are designed without considering the curse of dimensionality and stability.

Spencer et al. [24] demonstrate that the application of machine learning has significantly improved the accuracy of computer-aided diagnosis systems. Their study focuses on feature selection techniques for classification algorithms dependent on filters, using methods such as BayesNet, Logistic, Adaboost M1, JRip, Random Forest, and Stochastic Gradient Descent. They achieved 85% accuracy using a combination of BayesNet and Chi-squared, noting that variations in feature selection affect machine learning approaches.

Filter methods are crucial for feature selection as they can be used with any machine learning model and significantly shorten the execution time of algorithms. Bommert et al. [9] benchmarked selection approaches for high-dimensional datasets, examining 22 filter techniques using 16 high-dimensional classification datasets. Information theoretic, statistical, and random forest test filters performed well, though not every filter technique outperformed others in all scenarios. Their work serves as a reference for applications involving feature filtering.

Zacharias et al. [31] discuss the growing impact of AI systems on humans and the importance of transparency to uphold people's "right to explanation." They focus on explainable feature selection in AI systems, evaluating effectiveness, comprehension of feature impact, and justifiability. Their iterative design research cycle aims to create and assess artifacts to enhance the openness and comprehensibility of feature selection tasks.

Dimensionality reduction techniques in feature selection and feature extraction improve data mining and machine learning tasks by reducing redundancy, removing irrelevant data, and enhancing model accuracy and efficiency [32]. Techniques such as Principal Component Analysis (PCA) and t-SNE project features into a lower-dimensional space, preserving crucial data for analysis while improving processing and storage efficiency [15,21].

Enhancing classification performance across various domains requires careful consideration of feature selection for classification using PCA and Information Gain (IG). Research has shown that hybrid models integrating PCA and IG reduce data dimensions, select suitable features, and improve classification metrics such as accuracy, precision, and recall [19,17]. PCA significantly reduces preprocessing time while maintaining comparable performance outcomes, particularly in contexts like classifying Influenza-A antiviral resistance [23].

Explainable Artificial Intelligence (XAI) strategies, combined with feature selection methods, are critical for improving the interpretability and transparency of ML models, especially in the biomedical domain [28]. These techniques reduce the number of variables in model explanations and decision-making processes while preserving essential information [25]. Various machine learning models, including k-nearest neighbour, xgboost, Adaboost, logistic regression, random forest, decision tree, naive Bayes, and support vector machine, have been integrated with feature selection techniques like ReliefF to maximize performance [10].

Developing feature selection techniques grounded in explainable artificial intelligence (XAI) addresses the lack of interpretability and transparency in artificial neural networks, particularly in defect diagnostics systems [12]. For applications such as Intrusion Detection Systems (IDS), feature selection enhances both explainability and prediction performance in ML models [6]. In biological applications, balancing explainability, accuracy, and retention rate in feature selection has a substantial impact on final model explanations [28]. Researchers stress the importance of measuring causal strength and optimizing feature selection models for better interpretability and robustness using objective functions like relative entropy distance (RED).

3 Methodology

In this section, we outline the steps followed in our study. Our methodology is organized into several key stages: data collection, feature selection using principal component analysis (PCA), information gain, and explainable AI (XAI). This is followed by machine learning model selection and customization to interpret the model predictions. The proposed methodology is summarized in Figure 1, providing a clear and concise overview of our strategy.

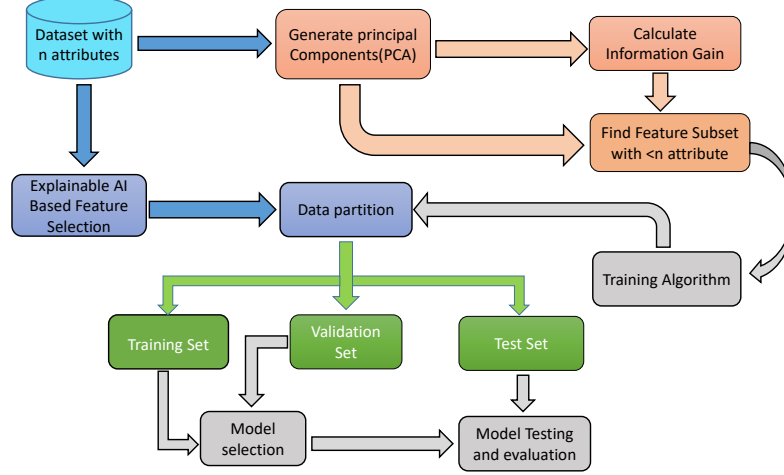


Fig. 1: Proposed Feature Selection Strategy for Imbalanced Data Classification.

3.1 Data Collection

The dataset utilized in this study, named "Imbalanced Data-based Prediction and Risk Factor Analysis of Stroke", is available on Mendeley[29]. It is based on data from the National Health and Nutrition Examination Survey and other national surveys. The dataset contains the records of 4,603 patients, with 362 (7.86%) diagnosed with stroke and 4,241 (92.14%) classified as non-stroke patients. It comprises 36 columns: the first column represents the stroke outcome variable, and the remaining columns serve as predictors. These predictors include gender, age, race, marital status, alcohol consumption, smoking status, sleep disorder, health insurance status, general health condition, depression, sleep time, diabetes, hypertension, high cholesterol, minutes of sedentary activity, coronary heart disease, body mass index (BMI), waist circumference, systolic blood pressure, diastolic blood pressure, high-density lipoprotein (HDL), triglycerides, low-density lipoprotein (LDL), fasting glucose, glycohemoglobin, energy intake, protein intake, carbohydrate intake, dietary fiber intake, total fat intake, total saturated fatty acids intake, total monounsaturated fatty acids intake, total polyunsaturated fatty acids intake, potassium intake, and sodium intake. This extensive dataset, which is entirely composed of numerical data, is used to study the prediction and analysis of stroke risk factors. We have three additional imbalanced datasets for testing our proposed method:

IT Customer Churn (20 features), Bank Marketing (17 features), and Car Insurance Claim (25 features), all available on Kaggle [26][13][14]. The IT Customer Churn dataset has 1,869 churn and 5,174 non-churn records. The Bank Marketing dataset includes 5,289 deposit and 5,873 non-deposit records. And the Car Insurance Claim dataset comprises 2,746 claim and 7,556 non-claim records. To ensure robust model performance, we divided the dataset into 80% training and 20% testing, with an additional 20% remained aside as a validation subset. Table 1 shows the data partitioned specifically for model training.

Table 1: Train, Test, and Validation split of the employed datasets.

Dataset Name	Target Column	Number of Samples	Train	Test	Valid
Imbalanced Stroke	stroke	4603	2946	921	736
IT Customer Churn	Churn	7043	4508	1409	1126
Bank Marketing	deposit	11162	7132	2232	1798
Car insurance claim	CLAIM_FLAG	10302	6593	2060	1649

3.2 Principal Component Analysis (PCA) for Feature Reduction

Due to the large number of features and high dimensionality in our imbalanced datasets, PCA is an effective method for data reduction. PCA is a dimensionality reduction approach that converts a collection of potentially correlated variables into a set of linearly uncorrelated variables known as principal components. PCA computes the covariance matrix of centered data, performs eigenvalue decomposition to obtain eigenvectors and eigenvalues, and selects principal components based on the largest eigenvalues.

Given a dataset X with N samples and d features, PCA aims to find a transformation matrix W such that,

$$X_{new} = X \cdot W \quad (1)$$

PCA converts the original data matrix X (size $n \times d$) into a reduced-dimensional space using W (size $d \times k$), where K is the number of principal components. The results X_{new} (size $n \times k$), captures the dataset's variance with the most significant orthogonal components.

3.3 Feature Selection Techniques

After performing PCA for feature reduction, we applied four different feature selection techniques. They are briefly described below.

Correlation-based Feature Selection Evaluation (CfsSubsetEval) CfsSubsetEval evaluates subsets of features based on their correlation with the target variable and the average correlation among features within the subset. Aiming to minimize redundancy among features, it selects subsets that maximize predictive accuracy. The merit $M(S)$ of a subset S of features is calculated as follows,

$$M(S) = \frac{k \cdot \text{cor}(\mathbf{X}_i, \mathbf{y})}{\sqrt{k + \frac{k \cdot (k-1)}{2} \cdot \text{avgCor}(\mathbf{X})}} \quad (2)$$

In CfsSubEval, K is the subset size, $\text{Cor}(X_i, y)$ is feature target correlation, and $\text{avgCor}(X)$ is the average feature correlation within S , crucial for feature selection.

Gain Ratio Attribute Evaluation (GainRatioAttributeEval) A feature selection technique called GainRatioAttributeEval assesses attributes according to their gain ratio, which quantifies how well an attribute classifies instances in relation to the inherent information of classes.

The gain ratio $GR(S)$ for a subset S of attributes is calculated as follows,

$$GR(S) = \frac{IG(S)}{IV(S)} \quad (3)$$

In GainRatioAttributeEval, $IG(S)$ is the information gain and $IV(S)$ is the intrinsic of subset S , crucial for attribute evaluation

ReliefF Attribute Evaluation (ReliefFAttributeEval) A feature selection method called ReliefFAttributeEval estimates an attribute's capacity to discriminate between examples that are close to one another in order to determine how important it is. It adjusts a weight vector for each attribute based on the difference between the chosen instance and its neighbors. The weight $W[A]$ of an attribute A is updated based on the following equation.

$$W[A] = W[A] - \frac{1}{m} \sum_{i=1}^m (\text{diff}(A, \mathbf{x}_i, \text{nearest-hit}) - \text{diff}(A, \mathbf{x}_i, \text{nearest-miss})) \quad (4)$$

Where $W[A]$ is the weight of attribute A , m is the number of iterations, and $\mathbf{x}_i$ is the sampled instance. The terms $\text{diff}(A, \mathbf{x}_i, \text{nearest-hit})$ and $\text{diff}(A, \mathbf{x}_i, \text{nearest-miss})$ denote differences with the nearest same-class and different-class neighbors, respectively.

Principal Component Analysis - Information Gain (PCA-IG) PCA-IG combines PCA with Information Gain to select the most relevant features from a high-dimensional dataset. After applying PCA, information gain is used on the transformed features to assess their relevance to the target variable. Equation (1) describes PCA, while Information Gain for a feature A is defined as,

$$IG(A) = H(Y) - H(Y|A) \quad (5)$$

Where $H(Y)$ is the entropy of the target variable and $H(Y|A)$ is the conditional entropy of the target variable given the feature A .

This hybrid approach is ideal for large, complex datasets where traditional feature selection methods struggle with high dimensionality.

3.4 Explainable AI (XAI)-Based Feature Selection

We utilized LIME as XAI in selecting the important features from the dataset. The approach is briefly described in the following section.

Local Interpretable Model-agnostic Explanations (LIME) LIME simplifies complex model behaviors into interpretable forms, offering local interpretability to explain specific predictions. It identified features with both positive and negative impacts and we utilized random forest-based LIME feature selection for analysis. The LIME explanation model is represented as follows,

$$\hat{g}(x) = \operatorname{argmin}_{g \in G} L(f, g, \pi_x) + \Omega(g) \quad (6)$$

In LIME, the explanation model $\hat{g}(x)$ approximates the local behavior of a complex model using a set G of potential models, often linear. A loss function $L(f, g, \pi_x)$ is employed to optimize the simplicity and interpretability of the perturbed explanation model.

3.5 Machine Learning Model Selection and Hyper-parameter Tuning

After performing PCA and feature selection, we identified top-performing machine learning algorithms: Random Forest Classifier (RFC), Support Vector Classifier (SVC), K-Nearest Neighbors (KNN), Logistic Regressor (LR), XGBoost (XGB) Classifier, and Extra Trees Classifier for classifying the results in our dataset. After selecting these ML models, we initially evaluated them using their default settings. Subsequently, we adjusted the hyperparameters to enhance the models' accuracy. Table 2 presents the hyperparameter configurations for each model.

Table 2: Hyperparameter settings for each machine learning model.

Model	Hyperparameters Setting
RFC	n_estimators: 100, 200, 300, max_depth: None, 10, 20, 30, min_samples_split: 2, 5, 10, min_samples_leaf: 1, 2, 4, max_features: 'auto', 'sqrt', 'log2'
SVC	C: 1.0, kernel: 'rbf', gamma: 'scale', degree: 3
KNN	n_neighbors: 5, weights: 'uniform', algorithm: 'auto', leaf_size: 30, p: 2
LR	penalty: 'l2', C: 1.0, solver: 'lbfgs', 'liblinear', 'saga', max_iter: 100
XGB Classifier	n_estimators: 100, random_state: 42
Extra Tress Classifier	n_estimators: 100, random_state: 42

In Table 2, RFC is tuned with hyperparameters like n_estimators (100, 200, 300), max_depth (None, 10, 20, 30), min_samples_split (2, 5, 10), min_samples_leaf (1, 2, 4), and max_features ('auto', 'sqrt', 'log2'). These parameters control the forest's complexity, affecting its ability to generalize and computational efficiency. SVC optimizes classification using key hyperparameters: C for regularization strength (default 1.0), kernel (default 'rbf'), gamma (default 'scale'), and degree (default 3 for 'poly' kernels). KNN classifier relies on a single hyperparameter, n_neighbors, which specifies the number of neighbors to consider during classification (default is 5). LR uses hyperparameters like penalty (default 'l2') for regularization type, C (default 1.0) for regularization

strength inversely proportional to regularization, and solver ('lbfgs', 'liblinear', 'saga') for optimization method. The number of trees to be constructed during the boosting process is specified by the `n_estimators` parameter, and the seed value is supplied by `random_state` to guarantee reproducibility of findings for the XGB and Extra Trees classifiers. When these parameters are tuned optimally, the model performs better, resulting in increased robustness and prediction accuracy across a range of applications.

3.6 Experimental Setup

We used Google Colab[2], a cloud-based tool, to build and refine machine learning models in Python. Our approach involved using PCA and information gain to feature reduction and selection, as well as leveraging XAI techniques such as LIME. This system supplied reliable computational resources and allowed for prolonged testing without requiring a lot of resources. Google Colab's collaborative capabilities improve team relations and research efficiency by allowing for seamless code and dataset sharing. All code was done in Python, with the Scikit Learn module for machine learning [1,3].

4 Experimental Results and Analysis

In this section, we present an analysis of the prediction outcomes from various models, employing both the full feature sets and the selected features identified through PCA, IG and XAI methods. A comprehensive examination of these results is detailed below.

4.1 Performance Comparison Using All Features (without feature selection)

These findings summarize the classification experiment performed on the dataset before to feature selection, and include metrics such as precision, recall, F1 score, and accuracy. Table 3 shows a full summary of these findings.

Table 3 presents the performance results of the best machine learning models across various datasets with all features considered. For the stroke dataset, RFC, SVC, and LR all achieved a precision, recall, and F1 score of 91.74% accuracy. In the IT customer dataset, LR achieved the highest accuracy at 80.19%. The bank marketing dataset saw the XGB classifier attaining the top accuracy of 83.47%. Lastly, the car insurance claim dataset was best modeled by the Extra Trees classifier, which reached an accuracy of 97.33%.

4.2 Performance After Feature Selection

Further experiments were conducted across all datasets using a variety of feature selection techniques. These methods included Cfs Subset Evaluation, Gain Ratio Attribute Evaluation, ReliefF Attribute Evaluation, PCA in combination with IG, and LIME-based feature selection. For each dataset, PCA was utilized with cross-validation and cumulative importance methods to identify crucial features. Here, we present detailed results for the imbalanced stroke dataset, where the number of selected features varied between nine (9) and twenty-two (22). Subsequently, the selected features were evaluated using the four classification algorithms mentioned earlier. The results of these experiments are presented in Table 4 below.

In Table 4, the results of four classification algorithms, each evaluated with five feature selection techniques, are presented. In KNN, ReliefF achieved the best precision and recall scores, around

Table 3: Performance of Classification Methods with All Features.

Dataset Name	Algorithm	Precision	Recall	F1-Score	Accuracy in (%)
Imbalanced Stroke	RFC	0.84	0.91	0.87	91.74
	SVC	0.84	0.91	0.87	91.74
	KNN	0.84	0.91	0.87	91.42
	LR	0.84	0.91	0.84	91.74
IT Customer Churn	RFC	0.65	0.45	0.54	79.20
	LR	0.65	0.54	0.59	80.19
	SVC	0.41	0.59	0.48	66.78
Bank Marketing	RFC	0.80	0.86	0.83	83.29
	XGB Classifier	0.81	0.85	0.83	83.47
	LR	0.79	0.76	0.77	78.63
Car insurance claim	KNN	0.29	0.44	0.35	55.70
	Extra Trees Classifier	0.96	0.93	0.94	97.33
	LR	0.50	0.002	0.004	73.07

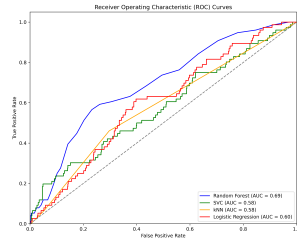
0.86 and 0.92, respectively. The combined PCA-IG method performed better in three classification models, with precision at 0.84, recall at 0.92, F1 score at 0.88, and an accuracy of 91.75%. LIME-based feature selection outperformed all other methods across various metrics. For SVC, KNN, and LR models, LIME and PCA-IG techniques produced similar results. However, LIME-based feature selection using a RFC achieved the highest overall performance, with a precision of 0.93, recall of 0.92, F1 score of 0.88, and accuracy of 91.86%. Overall, LIME-based feature selection with the RFC classifier provided the best results, indicating that XAI techniques are more effective than PCA in handling this imbalanced dataset.

The PCA-IG feature selection model resulted in 14 final features, while the LIME-based feature selection model identified 9 final features. These 9 features include 'Race', 'Alcohol', 'High Cholesterol', 'Coronary Heart Disease', 'Waist Circumference', 'Diastolic Blood Pressure', 'Total Fat', 'Total Saturated Fatty Acids', and 'Total Monounsaturated Fatty Acids'. Among them, 'Coronary Heart Disease', 'Total Saturated Fatty Acids', and 'Total Fat' are the top 3 positive contributors as identified by LIME.

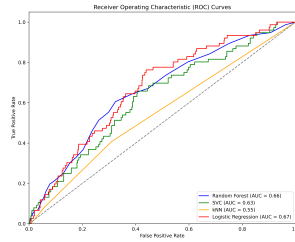
Figure 2 illustrates AUC curves for all classification models across three different feature selection scenarios in the imbalanced stroke dataset. In 2(a), utilizing full features, RFC had the greatest AUC of 0.69, followed by LR at 0.60. In 2(b), utilizing PCA-IG-selected features, LR had the greatest AUC of 0.67, whereas RFC scored 0.66. In 2(c), utilizing LIME-selected features, RFC and LR scored 0.61, whereas SVC and KNN performed similarly across scenarios. These findings show that each feature influences AUC performance, with PCA-IG and LIME-based selections generally producing better outcomes, albeit slightly lower overall AUC values.

Table 4: Performance of Classification Methods with Feature Selection in the Imbalanced Stroke Dataset.

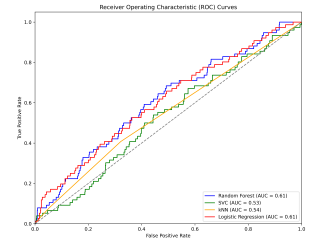
Classification Algorithm	Feature Selection	Precision	Recall	F1-Score	Accuracy in (%)
RFC	CfsSubsetEval	0.84	0.91	0.87	91.74
	GainRatioAttributeEval	0.84	0.91	0.87	91.74
	ReliefAttributeEval	0.84	0.91	0.87	91.74
	PCA-IG	0.84	0.91	0.88	91.75
	LIME	0.93	0.92	0.88	91.86
SVC	CfsSubsetEval	0.84	0.91	0.87	91.74
	GainRatioAttributeEval	0.84	0.91	0.87	91.74
	ReliefAttributeEval	0.84	0.91	0.87	91.74
	PCA-IG	0.84	0.92	0.88	91.75
	LIME	0.84	0.92	0.88	91.75
KNN	CfsSubsetEval	0.84	0.91	0.87	91.53
	GainRatioAttributeEval	0.84	0.90	0.87	91.66
	ReliefAttributeEval	0.86	0.92	0.87	91.53
	PCA-IG	0.84	0.91	0.87	91.09
	LIME	0.85	0.91	0.88	91.21
LR	CfsSubsetEval	0.84	0.91	0.87	91.74
	GainRatioAttributeEval	0.84	0.91	0.87	91.74
	ReliefAttributeEval	0.84	0.91	0.87	91.74
	PCA-IG	0.84	0.92	0.88	91.75
	LIME	0.84	0.92	0.88	91.75



(a) Full Features AUC Curves.



(b) PCA-IG Based AUC Curves.



(c) LIME Based AUC Curves.

Fig. 2: AUC Curve Comparison Across Multiple Classifier Models Using Three Feature Scenarios in the Imbalanced Stroke Dataset.

4.3 Testing of Feature Selection Methods on Different Datasets

Our proposed method, tested on an imbalanced stroke dataset, is now being evaluated on three additional imbalanced datasets: IT customer churn, bank marketing, and car insurance claim. Each dataset has a different best-performing ML model. Initially, we identify the best ML model using all features. Then, using LIME, we determine the most impactful features based on this model and examine the results accordingly. Table 5 provides detailed results of our analysis.

Table 5: Performance Analysis of Feature Selection Methods on Multiple Datasets.

Data set	ML Algorithm	Feature Selection	Precision	Recall	F1-Score	Accuracy (%)	AUC Curve
IT Customer Churn	LR	Full Features	0.65	0.54	0.59	80.19	0.85
		CfsSubSetEval	0.66	0.52	0.58	80.41	0.84
		GainRatioAttributeEval	0.62	0.30	0.40	76.65	0.81
		ReliefAttributeEval	0.64	0.52	0.57	79.84	0.84
		PCA-IG	0.69	0.52	0.59	81.05	0.84
		LIME	0.67	0.57	0.62	81.19	0.86
Bank Marketing	XGB Classifier	Full Features	0.81	0.85	0.83	83.47	0.91
		CfsSubSetEval	0.72	0.57	0.64	69.46	0.75
		GainRatioAttributeEval	0.72	0.45	0.56	65.65	0.70
		ReliefAttributeEval	0.81	0.85	0.83	83.78	0.90
		PCA-IG	0.81	0.85	0.83	83.21	0.91
		LIME	0.82	0.85	0.83	83.66	0.91
Car Insurance Claim	Extra Trees Classifier	Full Features	0.96	0.93	0.94	97.33	1.00
		CfsSubSetEval	0.99	0.98	0.99	99.56	1.00
		GainRatioAttributeEval	0.98	0.97	0.98	98.98	1.00
		ReliefAttributeEval	1.00	0.98	0.99	99.61	1.00
		PCA-IG	0.99	0.98	0.98	99.32	1.00
		LIME	0.99	0.99	0.99	99.66	1.00

In Table 5, the IT customer churn dataset shows the LR model performing the best, with an accuracy of 80.19% using all features. We then used PCA, IG, and XAI for feature selection. Both PCA-IG and LIME-based selections performed well. PCA-IG achieved a precision of 0.69 and accuracy of 81.05%. However, our proposed LIME-based method outperformed in all criteria, with

an accuracy of 81.19%. In the bank marketing dataset, the XGB classifier has the maximum accuracy across all features, scoring 83.47%. Using feature selection techniques, two models stand out: PCA-based ReliefAttributeEval and LIME-based selection. The ReliefAttributeEval model achieves an accuracy of 83.78%. However, our proposed LIME-based model outperforms the full feature model, achieving a precision of 0.82 and an accuracy of 83.66%. In the car insurance claim dataset, the Extra Trees classifier performs best with all features, achieving 97.33% accuracy. After feature selection, all methods improved accuracy, but our proposed LIME-based method outperformed them all. It achieved a precision, recall, and F1 score of 0.99, an accuracy of 99.66%.

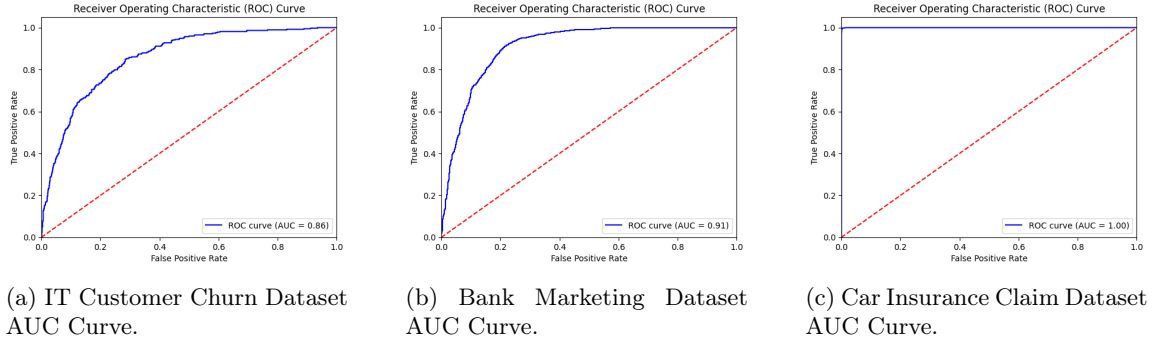


Fig. 3: AUC Curve for Best LIME-based Model Across Tested Datasets.

The AUC curves of the top-performing models with LIME-based feature selection are displayed in Figure 3. In Figure 3a, the Customer Churn dataset with an LR model achieved an AUC of 0.86. Figure 3b depicts the Bank Marketing dataset with an XGB classifier, achieving an AUC of 0.91. Figure 3c illustrates the Car Insurance Claim dataset with an Extra Trees classifier, reaching an AUC of 1.00. Across all datasets tested, our suggested LIME-based feature selection method consistently produced the best results for imbalanced datasets.

5 Discussion

This section discusses the results of our work, including experiments with various techniques and specifically using the LIME model with the RFC algorithm and other classification algorithms. When all features were used for classification, the algorithms yielded lower accuracy. This is expected, as including all data features can slow down machine learning algorithms due to the presence of potentially unnecessary data.

After feature selection and training, the PCA-IG and LIME models outperformed the other methods. Table 4 shows that PCA-IG performs better than basic PCA. However, LIME attained the maximum accuracy, with the PCA-IG model scoring 91.75% and the LIME model scoring 91.86% with the RFC classifier. We analyzed an imbalanced stroke dataset to determine the most influential factors affecting strokes. The LIME feature selection technique considers both positive and negative influences when determining the relevance of each feature. Following LIME-based feature selection, we identified nine features that had a positive impact on the model. The top three

contributions are 'Coronary Heart Disease', 'Total Saturated Fatty Acids', and 'Total Fat'. In the AUC study of the imbalanced stroke dataset, the full feature set yielded the highest AUC values. Feature selection with PCA, Information Gain, and LIME led to a slight drop in AUC scores but improved overall accuracy.

We also tested our results on three additional imbalanced datasets from different domains to verify our findings. In each case, the dataset's best ML model was identified first. Then, applying these ML models combined with LIME-based feature selection achieved the highest overall accuracy compared to using the full feature set, PCA, and PCA-IG techniques. Notably, the AUC scores improved with LIME-based feature selection across all tested datasets. This feature selection strategy enhances medical and other domain analyses by providing greater insights and aiding decision-making. By reducing data dimensions and selecting more meaningful features, we observed improvements in precision and recall, demonstrating that effective feature selection enhances machine learning efficiency. These results underscore the effectiveness of LIME in feature selection, leading to more accurate classifications.

6 Conclusion and Future Work

This study examined sophisticated feature selection methods with a focus on PCA, IG, and XAI (LIME) for the categorization of data that is imbalanced. Our experiments demonstrated that utilizing all features led to lower accuracy due to the inclusion of irrelevant data. Conversely, feature selection techniques such as PCA-IG and LIME successfully decreased the dimensionality of the data, improving the overall performance of the model and the relevance of the features that were chosen. Among these, LIME feature selection stood out, achieving the highest accuracy with the RFC on the imbalanced stroke dataset. In other tested datasets, the LIME-based feature selection technique resulted in the highest accuracy for the LR, XGB classifier, and Extra Trees classifier across various domain scenarios. These results illustrate the superiority of LIME in identifying and utilizing influential features, and they also highlight the promise of XAI approaches in enhancing classification outcomes for imbalanced datasets.

Future research will look into additional feature selection approaches and combinations to improve classification performance. Genetic algorithms, mutual information, and ensemble approaches will be discussed. Furthermore, using imbalanced data management strategies like SMOTE, ADASYN, and ensemble learning methods may improve model robustness. Another focus will be on investigating the applicability of these techniques in various domains and the use of deep learning technologies. The goal is to provide more adaptable and efficient feature selection frameworks for a variety of classification jobs.

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